📘 MODULE 24: HTTP EVENT COLLECTOR (HEC)

24.1 Concept

HEC lets applications send events to Splunk over **HTTP/HTTPS** with a **token**, with no forwarder needed.

24.2 Why it exists

Modern apps, containers, serverless functions, and scripts can't always install a UF. HEC gives them a simple, scalable, token-authenticated endpoint.

24.3 Key facts

- Default port **8088**, **HTTPS by default**.
- Authentication: header `Authorization: Splunk <token>`.
- A **token** is bound to settings: allowed indexes, default index, default sourcetype.
- Common endpoints:
  - `/services/collector/event`: JSON events
  - `/services/collector/raw`: raw text (needs a channel parameter in many setups)
  - `/services/collector/health`: health check
- **Splunk Cloud uses different URLs and port rules**, so check your stack's docs.

24.4 Enable and create a token

1. Settings → Data inputs → **HTTP Event Collector**.
2. **Global Settings:** All Tokens = **Enabled**, port `8088`, SSL enabled. Save.
3. **New Token:** name `hec_lab` → Next.
4. Source type: choose or set `hec_test`. **Allowed indexes:** `main` (and others you need). **Default index:** `main`. Review → Submit.
5. **Copy the token value** (a UUID).

24.5 Test it (run in Ubuntu terminal)

Health check:

```bash
curl -k https://localhost:8088/services/collector/health
```

- `curl` sends an HTTP request. `-k` skips certificate validation (self-signed cert in a lab only).
- Expected: `{"text":"HEC is healthy","code":17}`.

Send an event:

```bash
```

```
curl -k https://localhost:8088/services/collector/event \
  -H "Authorization: Splunk <your-token>" \
  -d '{"event": {"user":"alice","action":"login","status":"success"}, "sourcetype":"hec_te
```

- `-H` adds the auth header. `-d` is the request body (JSON).
- Expected: `{"text":"Success","code":0}`.

Search:

```spl
index=main sourcetype=hec_test | table _time host user action status
```

Because `event` was a JSON object, Splunk extracts `user`, `action`, `status` automatically at search time (JSON sourcetype behavior; if not, set `KV_MODE=json` for the sourcetype).

24.6 JSON event format

Key	Meaning
`event`	**Required.** String or object
`time`	Epoch time (seconds, decimals allowed). Omitted = receive time
`host`, `source`, `sourcetype`, `index`	Metadata overrides
`fields`	Indexed extra fields (use sparingly)

Batch: send several JSON objects one after another in the same request body.

24.7 Common errors

Response/symptom	Cause
Connection refused	HEC disabled globally, wrong port, Splunk down
Invalid token / 403	Wrong token, disabled token, malformed header (`Splunk <token>`)
Invalid index / `Incorrect index`	Index not in the token's **allowed indexes**, or doesn't exist
`No data` / bad JSON	Missing `event` key or malformed JSON
SSL error	HTTP used against HTTPS; use `https://` and `-k` for self-signed labs
Events missing	Wrong time range (bad `time` value), or searching a different index

24.8 Interview Q&A

- **What is HEC?** A token-authenticated HTTP(S) endpoint to send events directly to Splunk.
- **Default port?** 8088, HTTPS by default.
- **How does HEC authenticate?** A token in the `Authorization: Splunk <token>` header.
- **HEC vs UF?** HEC is for apps pushing data over HTTP, UF is for reading files and OS logs.
- **HEC returns success but data is missing. Why?** Wrong index or time, or a different index than expected. Check the token's default and allowed indexes.
- **Event vs raw endpoint?** `/event` takes JSON with metadata, `/raw` takes unstructured text.

24.9 Revision

HEC = token + HTTPS 8088 · `Authorization: Splunk <token>` · `/services/collector/event` · allowed indexes matter · `/health` to test.